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COOKING APPLIANCE AND METHOD FOR INITIALIZING CONTROLLER TERMS FOR COOKING OPERATIONS

Abstract

A cooking appliance includes at least one heating element; a temperature sensor; and a controller operably connected with the at least one heating element and the temperature sensor, the controller configured to perform a heating operation, the heating operation including determining that a temperature rate of change at the temperature sensor has reached a threshold rate of change; determining a threshold time point at which the temperature rate of change reaches the threshold rate of change; and determining one or more parameters for a feedback controlled cooking phase according to the determined threshold time point, the one or more parameters comprising at least one initial controller term value.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present subject matter relates generally to cooking appliances, and more particularly to methods of operating cooking appliances according to thermal behaviors of cookware items.

BACKGROUND OF THE INVENTION

[0002] Cooking appliances generally have one or more heating elements configured for heating a cookware item. The cookware item, e.g., a pot or a pan, may be positioned on or near the one or more heating elements and food products (including, e.g., food solids, liquid, or water) may be placed inside the cookware item for cooking. A controller may selectively energize the heating element(s) to provide thermal energy to the cookware item and the food products placed therein. Alternatively, certain cooking appliances, often referred to as induction cooktops, provide energy in the form of an alternating magnetic field which causes the cookware item to generate heat. In both types of appliances, a controller selectively energizes either the heating element(s) or a magnetic coil to heat the food products until they are properly cooked.

[0003] Cookware items may exhibit different thermal properties or behaviors. For instance, some cookware items may have slower heat transfer rates, retain heat more easily, or dissipate heat more easily. For cooking appliances that are capable of performing feedback controlled heating operations, one or more algorithms may be used to incorporate certain feedback information (e.g., temperature change, temperature rate of change, etc.) over a heating period to intelligently control a power level of the heating element(s). For instance, some heating operations may incorporate constant or fixed preheating phases during which the heating element is driven at a constant power level. When the cooking phase begins, controller terms (e.g., derivative, integral, etc.) may be set to institute the feedback controlled operation. However, existing methods have several drawbacks, including inadequate controller output, large overshoots, and the like.

[0004] Accordingly, a cooking appliance and method of operating a cooking appliance that obviates one or more of the above-mentioned drawbacks would be desirable. In particular, a cooking appliance capable of adjusting one or more initial parameters of a heating operation would be useful.

BRIEF DESCRIPTION OF THE INVENTION

[0005] Aspects and advantages of the invention will be set forth in part in the following description, or may be obvious from the description, or may be learned through practice of the invention.

[0006] In one exemplary aspect of the present disclosure, a cooking appliance is provided. The cooking appliance may include at least one heating element to selectively supply heat to a cookware item; a temperature sensor configured to selectively monitor a temperature of the cookware item; and a controller operably connected with the at least one heating element and the temperature sensor, the controller configured to perform a heating operation. The heating operation may include determining a temperature setpoint; directing the at least one heating element based on the determined temperature setpoint over a preheating phase; monitoring a temperature rate of change at the temperature sensor over the preheating phase; determining that the temperature rate of change has reached a threshold rate of change; determining a threshold time point at which the temperature rate of change reaches the threshold rate of change; determining one or more parameters for a feedback controlled cooking phase according to the determined threshold time point, the one or more parameters including at least one initial controller term value; and directing the at least one heating element according to the one or more determined parameters for a duration of the feedback controlled cooking phase.

[0007] In another exemplary aspect of the present disclosure, a method of operating a cooking appliance is provided. The cooking appliance may include at least one heating element and a temperature sensor. The method may include determining a temperature setpoint; directing the at least one heating element based on the determined temperature setpoint over a preheating phase;

monitoring a temperature rate of change at the temperature sensor over the preheating phase; determining that the temperature rate of change has reached a threshold rate of change; determining a threshold time point at which the temperature rate of change reaches the threshold rate of change; determining one or more parameters for a feedback controlled cooking phase according to the determined threshold time point, the one or more parameters including at least one initial controller term value; and directing the at least one heating element according to the one or more determined parameters for a duration of the feedback controlled cooking phase.

[0008] These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

[0010] FIG. 1 provides a perspective view of an oven range according to exemplary embodiments of the present disclosure.

[0011] FIG. 2 provides a side cut-away view of the exemplary oven range of FIG. 1.

[0012] FIG. 3 provides a graph illustrating a comparison of a cookware temperature, a sensor temperature, and controller terms over time for a cookware item according to exemplary embodiments of the present disclosure.

[0013] FIG. 4 provides a table illustrating a plurality of controller term values for a plurality of sensor times to a temperature rate of change threshold according to exemplary embodiments of the present disclosure.

[0014] FIG. 5 provides a graph illustrating a sensor temperature change over time of an exemplary heating operation.

[0015] FIG. 6 provides a graph illustrating multiple sensor temperature rates of change of different cookware items during exemplary heating operations.

[0016] FIG. 7 provides a flow chart illustrating a method of operating a cooking appliance according to exemplary embodiments of the present disclosure.

[0017] Repeat use of reference characters in the present specification and drawings is intended to represent the same or analogous features or elements of the present invention.

DETAILED DESCRIPTION

[0018] Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0019] As used herein, the terms “first,” “second,” and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components. The terms “includes” and “including” are intended to be inclusive in a manner similar to the term “comprising.” Similarly, the term “or” is generally intended to be

inclusive (i.e., “A or B” is intended to mean “A or B or both”). In addition, here and throughout the specification and claims, range limitations may be combined and/or interchanged. Such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other. The singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

[0020] Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “generally,” “about,” “approximately,” and “substantially,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components and/or systems. For example, the approximating language may refer to being within a 10 percent margin, i.e., including values within ten percent greater or less than the stated value. In this regard, for example, when used in the context of an angle or direction, such terms include within ten degrees greater or less than the stated angle or direction, e.g., “generally vertical” includes forming an angle of up to ten degrees in any direction, e.g., clockwise or counterclockwise, with the vertical direction V.

[0021] The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” In addition, references to “an embodiment” or “one embodiment” does not necessarily refer to the same embodiment, although it may. Any implementation described herein as “exemplary” or “an embodiment” is not necessarily to be construed as preferred or advantageous over other implementations. Moreover, each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

[0022] FIG. 1 provides a perspective view of a cooking appliance, or oven range **10**, including a cooktop **12**, and FIG. 2 provides a side cut-away view of the cooking appliance **10**. Cooking appliance **10** is provided by way of example only and is not intended to limit the present subject matter to the arrangement shown in FIGS. 1 and 2. Thus, the present subject matter may be used with other range **10** and/or cooktop **12** configurations, e.g., double oven range appliances. As illustrated, cooking appliance **10** generally defines a vertical direction V, a lateral direction L, and a transverse direction T, each of which is mutually perpendicular, such that an orthogonal coordinate system is generally defined. Cooking appliance **10** may include a cabinet **101** that extends between a top **103** and a bottom **105** along the vertical direction V, between a left side **107** and a right side **109** along the lateral direction, and between a front **111** and a rear **113** along the transverse direction T.

[0023] A cooking surface **14** of cooktop **12** may include a plurality of heating elements **16**. For the embodiment depicted, cooktop **12** includes five heating elements **16** spaced along cooking surface **14**. Heating elements **16** may be electric heating elements and are positioned at, e.g., on or proximate to, the cooking surface **14**. In certain exemplary embodiments, cooktop **12** is a radiant cooktop with resistive heating elements or coils mounted below cooking surface **14**. However, in other embodiments, the cooktop appliance **12** includes other suitable shape, configuration, and/or number of heating elements **16**, for example, cooktop **12** may be an open coil cooktop with heating elements **16** positioned on or above surface **14**. Additionally or alternatively, in other embodiments, cooktop **12** may include any other suitable type of heating element **16**, such as an induction heating element. Each of the heating elements **16** may be the same type of heating element **16**, or cooktop

12 may include a combination of different types of heating elements **16**.

[0024] As mentioned, heating element **16** may be an induction style heating element. Thus, as would be understood by those skilled in the art, appliance **10** may supply a current to heating element **16** (e.g., such as a Lenz coil). As such, current may pass through heating element **16** to generate a magnetic field. The magnetic field may be a high frequency circulating magnetic field. The magnetic field may be directed towards and through cooktop appliance **12** to a cookware item (e.g., cookware item **18**, described below). In particular, when the magnetic field penetrates cookware item **18**, the magnetic field induces a circulating electrical current within cookware item **18**. The material properties of cookware item **18** may restrict a flow of the induced electrical current and convert the induced electrical current into heat within cookware item **18**. As cookware item **18** heats up, contents of cookware item **18** contained therein heat up as well. In such a manner, the induction heating element can cook the contents of cookware item **18**.

[0025] As shown in FIG. **1**, a cooking utensil (or cookware item) **18**, such as a pot, pan, or the like, may be placed on a heating element **16** to heat cookware item **18** and cook or heat food items placed within cookware item **18**. Cooking appliance **10** may also include a door **20** that permits access to a cooking chamber **104** of oven range **10**, e.g., for cooking or baking of food items therein. A control panel **22** having controls **24** may permit a user to make selections for cooking of food items. Although shown on a backsplash or back panel **26** of oven range **10**, control panel **22** may be positioned in any suitable location.

[0026] Controls **24** may include buttons, knobs, and the like, as well as combinations thereof, and/or controls **24** may be implemented on a remote user interface device such as a smartphone. As an example, a user may manipulate one or more controls **24** to select a temperature and/or a heat or power output for each heating element **16** and the cooking chamber **104**. The selected temperature or heat output of heating element **16** affects the heat transferred to cookware item **18** placed on heating element **16**. A display **28** may be provided (e.g., on or in control panel **22**). Display **28** may display information regarding cooking operations or inputs from a user regarding the cooking operation. Display **28** may be any suitable display capable of providing visual feedback, such as a liquid crystal display (LCD), a light emitting diode (LED) display, a segmented display, or the like. Additionally or alternatively, display **28** may be a touch display capable of receiving touch inputs from a user.

[0027] Cooktop appliance **12** may further include or be in operative communication with a processing device or a controller **50** that may be generally configured to facilitate appliance operation. In this regard, control panel **22**, controls **24**, and display **28** may be in communication with controller **50** such that controller **50** may receive control inputs from controls **24**, may display information using display **28**, and may otherwise regulate operation of cooking appliance **10**. For example, signals generated by controller **50** may operate cooking appliance **10**, including any or all system components, subsystems, or interconnected devices, in response to the position of controls **24** and other control commands. Control panel **22** and other components of appliance **10** may be in communication with controller **50** via, for example, one or more signal lines or shared communication busses. In this manner, Input/Output (“I/O”) signals may be routed between controller **50** and various operational components of appliance **10**.

[0028] As used herein, the terms “processing device,” “computing device,” “controller,” or the like may generally refer to any suitable processing device, such as a general or special purpose microprocessor, a microcontroller, an integrated circuit, an application specific integrated circuit (ASIC), a digital signal processor (DSP), a field-programmable gate array (FPGA), a logic device, one or more central processing units (CPUs), a graphics processing units (GPUs), processing units performing other specialized calculations, semiconductor devices, etc. In addition, these “controllers” are not necessarily restricted to a single element but may include any suitable number, type, and configuration of processing devices integrated in any suitable manner to facilitate appliance operation. Alternatively, controller **50** may be constructed without using a

microprocessor, e.g., using a combination of discrete analog and/or digital logic circuitry (such as switches, amplifiers, integrators, comparators, flip-flops, AND/OR gates, and the like) to perform control functionality instead of relying upon software.

[0029] Controller **50** may include, or be associated with, one or more memory elements or non-transitory computer-readable storage mediums, such as RAM, ROM, EEPROM, EPROM, flash memory devices, magnetic disks, or other suitable memory devices (including combinations thereof). These memory devices may be a separate component from the processor or may be included onboard within the processor. In addition, these memory devices can store information and/or data accessible by the one or more processors, including instructions that can be executed by the one or more processors. It should be appreciated that the instructions can be software written in any suitable programming language or can be implemented in hardware. Additionally, or alternatively, the instructions can be executed logically and/or virtually using separate threads on one or more processors.

[0030] For example, controller **50** may be operable to execute programming instructions or micro-control code associated with an operating cycle of cooking appliance **10**. In this regard, the instructions may be software or any set of instructions that when executed by the processing device, cause the processing device to perform operations, such as running one or more software applications, displaying a user interface, receiving user input, processing user input, etc. Moreover, it should be noted that controller **50** as disclosed herein is capable of and may be operable to perform any methods, method steps, or portions of methods as disclosed herein. For example, in some embodiments, methods disclosed herein may be embodied in programming instructions stored in the memory and executed by controller **50**.

[0031] The memory devices may also store data that can be retrieved, manipulated, created, or stored by the one or more processors or portions of controller **50**. The data can include, for instance, data to facilitate performance of methods described herein. The data can be stored locally (e.g., on controller **50**) in one or more databases and/or may be split up so that the data is stored in multiple locations. In addition, or alternatively, the one or more database(s) can be connected to controller **50** through any suitable network(s), such as through a high bandwidth local area network (LAN) or wide area network (WAN). In this regard, for example, controller **50** may further include a communication module or interface that may be used to communicate with one or more other component(s) of appliance **10**, controller **50**, an external appliance controller, or any other suitable device, e.g., via any suitable communication lines or network(s) and using any suitable communication protocol. The communication interface can include any suitable components for interfacing with one or more network(s), including for example, transmitters, receivers, ports, controllers, antennas, or other suitable components.

[0032] Cooking appliance **10** may include a temperature sensor **40**. Temperature sensor **40** may be configured to selectively sense a temperature of a cookware item (e.g., cookware item **18**) as it is heated. For instance, temperature sensor **40** may be integrally formed with cooking appliance **10** (e.g., within cooktop **12**, within cooking chamber **104**, etc.). In some embodiments, temperature sensor **40** is operably connected to cooking appliance **10** (e.g., via a port or socket, via a remote connection, etc.). For one example, temperature sensor **40** is provided within cookware item **18** and operably connected to controller **50** during a cooking operation. Temperature sensor **40** may monitor a temperature of cookware item **18** or a food item provided within cookware item **18**. Accordingly, temperature sensor **40** may deliver signals (e.g., voltage signals) representing the temperature of cookware item **18** to controller **50**. The signals may be sent according to a predetermined frequency (e.g., at predetermined time intervals). Thus, controller **50** may analyze a temperature or temperature change of cookware item **18**.

[0033] As used herein, “temperature sensor” or the equivalent is intended to refer to any suitable type of temperature measuring system or device positioned at any suitable location for measuring the desired temperature. Thus, for example, temperature sensor **40** may be any suitable type of

temperature sensor, such as a thermistor, a thermocouple, a resistance temperature detector, a semiconductor-based integrated circuit temperature sensor, etc. In addition, temperature sensor **40** may be positioned at any suitable location and may output a signal, such as a voltage, to a controller that is proportional to or indicative of the temperature being measured. Although exemplary positioning of temperature sensors is described herein, it should be appreciated that appliance **10** may include any other suitable number, type, and position of temperature or other sensors according to alternative embodiments.

[0034] FIG. **3** provides a graph illustrating a cookware item temperature and setpoint, a sensor temperature and setpoint, an integral term, a derivative term, and a total proportion-integral-derivative (PID) output value for an exemplary cookware item. It should be noted that a proportional term is also incorporated into the PID output value but not shown individually in the graph. In detail, as mentioned above, temperature sensor **40** may monitor the temperature of cookware item **18** over the course of the cooking or heating operation. The heating operation may include a preheating phase and a cooking phase. The preheating phase may be a constant heating preheating phase. For instance, during the preheating phase, heating element **16** may be driven (e.g., powered) at a constant predetermined power level for a duration of the preheating phase. Accordingly, a controlled temperature increase may be exhibited by cookware item **18**.

[0035] During the preheating phase, a temperature rate of change (ROC) at temperature sensor **40** may be monitored. Temperature sensor **40** may continually monitor a temperature throughout the preheating phase. For instance, the ROC may be determined as a change in temperature (e.g., ΔT) over a change in time (e.g., Δt). Thus, a difference between the end temperature sensed at temperature sensor **40** (e.g., in degrees Fahrenheit) and the starting or initial temperature at temperature sensor **40** may be divided by a total length of a time window (e.g., in seconds) to determine an instantaneous ROC. For instance, as will be explained in more detail below, the temperature ROC may be monitored over a fixed-length rolling time window. The fixed-length rolling time window may have a predetermined advancement period. For instance, the predetermined advancement period may be between about 0.5 second and about 1.5 seconds.

[0036] Referring briefly to FIG. **5**, a change in temperature at the temperature sensor over time (e.g., over the preheating phase) is illustrated. As shown, the temperature may increase as the preheating phase progresses. The fixed-length rolling time window may be initiated at or near an initiation of the preheating phase. For instance, a first temperature $T_{sub.A}$ may be taken at a first time point $t_{sub.A}$. The first temperature $T_{sub.A}$ may thus be taken at or near the initiation of the preheating phase. A second temperature $T_{sub.B}$ may be taken at a second time point $t_{sub.B}$. The second time point $t_{sub.B}$ may be at the end of the fixed-length rolling time window. For at least one example, the time window is between about 10 seconds and about 30 seconds, or about 20 seconds. Accordingly, the first and second temperatures $T_{sub.A}$ and $T_{sub.B}$ may be taken approximately 20 seconds apart from each other. It should be understood that the length of the time window may vary according to specific embodiments, and the disclosure is not limited to the examples provided herein.

[0037] Additionally or alternatively, as mentioned, the time window may be a rolling time window with the predetermined advancement period. Accordingly, temperature measurements may be taken at regular intervals over the preheating phase. While determining the ROC, temperatures that are spaced apart from each other by the length of the time window may be compared. The difference between the two temperatures (e.g., $T_{sub.A}$ and $T_{sub.B}$) may then be divided by the length of the time period (e.g., about 20 seconds) to determine the ROC (e.g., as determined at the second time point $t_{sub.B}$). For instance, in some embodiments, the ROC may be subjected to a smoothing or noise reduction operation (e.g., a moving average, exponential smoothing, etc.).

[0038] As the time window is a rolling time window, multiple temperature measurements may be made and stored (e.g., temporarily) within a memory to be compared against future temperatures. For instance, a first time window may overlap a second time window. According to this instance, a

first time window may include a first time point t.sub.A1 and a second time point t.sub.B1. Likewise, the second time window may include a first time point t.sub.A2 and a second time point t.sub.B2. Time point t.sub.A2 may be between time point t.sub.A1 and time point t.sub.B1. Accordingly, time point t.sub.B2 may be after time point t.sub.B1. As such, the ROC is determined as a rolling ROC having multiple time windows of fixed time length.

[0039] As shown in FIG. 6, multiple different cookware items may cause the temperature at temperature sensor 40 to exhibit multiple different temperature ROCs over the preheating phase. For instance, multiple different aspects of cookware item 18 may result in different thermal behaviors, such as pan size, density, material composition, weight, surface contact area, etc. Accordingly, each cookware item may absorb, transmit, or otherwise transfer heat differently. As shown in FIG. 6, the thermal behavior of the cookware item may be deduced according to the temperature ROC (e.g., at temperature sensor 40). For instance, the ROC (or smoothed ROC) may be compared against a temperature ROC threshold. The cookware item may then be, e.g., classified according to an amount of time taken for the ROC to reach or cross the ROC threshold. As heat and power are added to the system (e.g., cookware item 18 or temperature sensor 40), the temperature ROC at temperature sensor 40 may be expected to increase. Accordingly, the temperature ROC threshold may be a positive number (e.g., greater than zero).

[0040] The cooking phase may be a feedback controlled cooking phase. In detail, the cooking phase may intelligently adjust one or more parameters according to feedback with respect to cookware item 18, a food being cooked, cooking appliance 10, or the like. Temperature sensor 40 may continually send temperature signals to controller 50 which may then determine, for instance, an error value associated with the feedback controlled cooking phase. The error value may be a difference between a temperature setpoint and an actual observed temperature (e.g., via temperature sensor 40). The error value may be substituted into a feedback equation to determine an adjustment to be made to a control variable. For instance, the control variable may be a power level of heating element 16.

[0041] According to at least some embodiments, controller 50 includes a closed-loop feedback control algorithm. The closed-loop feedback control algorithm may be a proportional-integral-derivative (PID) algorithm or equation (e.g., equation or set of equations). In some embodiments, the algorithm may include a proportional algorithm, a proportional-integral algorithm, a proportional-derivative algorithm, or any suitable combination of terms. The PID controller may determine a proportional term (P), an integral term (I), and a derivative term (D). The PID algorithm may be:

[00001] $CV = P + I + D$ [0042] where CV is a controlled variable (e.g., power input to heating element 16), P is the proportional term, I is the integral term, and D is the derivative term. As can be seen, adding each of the P, I, and D terms generates a value for the power level of heating element 16. Each of the P, I, and D terms may be found as follows:

[00002] $P = K_p * e$ $I = I_{prev} + K_i * e * T_s$ $D = K_d * (e - e_{prev}) / T_s$ [0043] where K.sub.p is a proportional gain value, K.sub.i is an integral gain value, K.sub.d is a derivative gain value, e is an error value (e.g., a difference between a temperature setpoint and an observed temperature), T.sub.s is a sampling time or sampling time rate (e.g., a rate at which a discrete system samples inputs), I.sub.prev is a previous integral term (e.g., at the previous sampling event), and e.sub.prev is a previous error value (e.g., at the previous sampling event). As noted above, however, in some instances any suitable combination of P, I, and D terms may be incorporated into the algorithm.

[0044] In some instances, the derivative (D) term may be susceptible to high levels of noise. Thus, large oscillations of the D term may be observed throughout the feedback controlled cooking phase. Accordingly, the D term may be subjected to a filtering to reduce the noise and obtain a more steady, predictable term over the cooking phase. For one example:

[00003] $D_{filtered} = * D + (1 -) * D_{filtered_{prev}}$ [0045] where D.sub.filtered is the filtered D term,

D.sub.filteredprev is the previous filtered D term (e.g., from a measuring point immediately preceding the current measuring point [previous sampling event]), and a is a filter smoothing factor, such that $0 \leq a \leq 1$. As would be understood, a smaller value of a (e.g., closer to 0) would result in greater smoothing of the D term. Additionally or alternatively, the filtered D term incorporates previous D terms to provide smoother adjustments to the D term of the PID controller algorithm. With a smoother D term, fluctuations of the PID controller outputs may be reduced. Thus, the PID algorithm may be adjusted to:

$$[00004] CV = P + I + D_{\text{filtered}}$$

[0046] According to some instances, each of the I term and the D term may be initialized (e.g., to a non-zero value) at the completion of the preheating phase before initiating the feedback controlled cooking phase. As shown in FIG. 3, for instance, the feedback controlled cooking phase may initiate at 2.5 minutes (e.g., at the conclusion of the preheating phase). At the initiation of the cooking phase, the I term and the D term are each initialized to a non-zero value. According to the example provided herein, the I term is initialized to a positive value while the D term is initialized to a negative value. It should be understood, however, that each of the I term and the D term may be initialized to any suitable value, and the disclosure is not limited to the examples provided herein.

[0047] FIG. 4 provides a table illustrating a plurality of initialized values according to an amount of time for the sensor temperature to reach a predetermined rate of change. As mentioned above, the temperature ROC for cookware item **18** (e.g., at temperature sensor **40**) may be monitored over the course of the preheating phase. The initialized I term and D term may be retrieved according to when the temperature ROC at temperature sensor **40** crosses the threshold ROC during the preheating phase. For instance, cooking appliance **10** may incorporate a method to determine the initialized I term and D term.

[0048] Now that the construction of cooking appliance **10** and a configuration of controller **50** according to exemplary embodiments have been presented, an exemplary method **300** of operating a cooking appliance will be described. Although the discussion below refers to the exemplary method **300** of operating cooking appliance **10**, one skilled in the art will appreciate that the exemplary method **300** is applicable to the operation of a variety of other cooking appliances. In exemplary embodiments, the various method steps as disclosed herein may be performed by controller **50** or a separate, dedicated controller. Additionally or alternatively, the various method steps may be performed in a different order, including additional steps or omitting certain steps according to specific embodiments.

[0049] At step **302**, method **300** may include determining a temperature setpoint. In detail, a user may communicate with the cooking appliance (e.g., cooking appliance **10**) a desire to initiate a cooking operation, a heating operation, or the like. For example, the cooking operation may include a feedback controlled heating phase incorporating a PID algorithm to continually monitor the heating operation and perform adjustments as needed. For instance, as will be discussed, the cooking operation may include a preheating phase before the heating phase. According to at least some embodiments, the preheating phase may not incorporate feedback control (e.g., PID feedback control). The user may manually enter a temperature setpoint (e.g., a temperature at which the user desires to have the item cooked). Thus, using a user interface (e.g., control panel **22**), the user may enter a specific cooking temperature as the temperature setpoint (e.g., 250° F., 300° F., 350° F., etc.). In additional or alternative embodiments, the user may provide information regarding a specific food item to be cooked (e.g., eggs, meat, vegetables, etc.). For instance, the cooking appliance may include features for selecting predetermined food items from the user interface or the cooking appliance may include a remote connectivity (e.g., wireless fidelity [WiFi], Bluetooth®, etc.), through which the user may select a food item (e.g., via a remote device). Further still, the user may input a particular recipe to be cooked on or in the cooking appliance. The temperature setpoint may be stored within the cooking appliance (e.g., within a controller or a memory therein).

[0050] At step **304**, method **300** may include directing the at least one heating element based on the determined setpoint over a preheat or preheating phase. As mentioned above, the cooking or heating operation may include each of the preheating phase and the cooking phase. The preheating phase may include operating the at least one heating element at a constant power level over a duration of the preheating phase. For instance, method **300** may include determining the power level for the at least one heating element based on the determined temperature setpoint.

Accordingly, the power level of the at least one heating element may vary based on the desired cooking temperature, the food being cooked, a particular recipe, or the like. Accordingly, the at least one heating element may then be directed at the determined power level for the duration of the preheating phase.

[0051] According to some embodiments, a total duration of the preheating phase may be predetermined. For instance, the preheating phase may be set to a fixed time length. In additional or alternative embodiments, the preheating phase may be determined to be completed when a temperature at the temperature sensor reaches a sensor temperature setpoint (e.g., target temperature). The sensor temperature setpoint may be different from the determined setpoint. For instance, the sensor temperature setpoint or target may be based on the determined setpoint mentioned above.

[0052] At step **306**, method **300** may include monitoring a temperature rate of change (ROC) at the temperature sensor over the preheating phase. As described above, the temperature at the temperature sensor may be regularly measured throughout the preheating phase. For instance, the temperature sensor may determine and send (e.g., to a controller) temperature signals at multiple discrete time points throughout the preheating phase. As mentioned, the temperature ROC may be monitored over a fixed-length rolling time window. The fixed-length time window may be between about 10 seconds and about 30 seconds. Additionally or alternatively, the fixed-length rolling time window may have a predetermined advancement period. The predetermined advancement period may be between about 0.5 second and 1.5 seconds. Thus, the temperature ROC may be continually monitored over the preheating phase.

[0053] At step **308**, method **300** may include determining that the temperature ROC has reached a threshold ROC. The threshold ROC may be a predetermined ROC at which the cookware item may be, e.g., classified or determined. The threshold ROC may be stored within the appliance or a remote connected device. Additionally or alternatively, the threshold ROC may vary according to the determined temperature setpoint. For instance, a plurality of threshold ROCs may be stored and retrieved based on the temperature setpoint.

[0054] At step **310**, method **300** may include determining a threshold time point at which the ROC reaches the threshold ROC. As mentioned, the temperature measurements may be taken at distinct time points. For an exemplary time window, a first temperature may be taken at an **A1** time point and a second temperature may be taken at a **B1** time point. In the event that the temperature ROC has reached the threshold ROC at the conclusion of the exemplary time window, the **B1** time point may be noted as the threshold time point. Method **300** may calculate a length of time from an initiation of the preheating phase to the threshold time point (e.g., **B1**). For instance, a timer may be initiated at the start of the preheating phase. The timer may then be stopped at the point at which the temperature ROC at the temperature sensor reaches or surpasses the threshold ROC.

[0055] At step **312**, method **300** may include determining one or more parameters for a feedback controlled cooking phase according to the determined threshold time point. For instance, as described above, the feedback controlled cooking phase may incorporate a PID control algorithm utilizing a proportional term (P), an integral term (I), and a derivative term (D). Thus, the one or more parameters for the feedback controlled cooking phase may include an initial term value for at least one of the I term or the D term. According to some embodiments, the initial term value or values may be retrieved from a lookup table (e.g., as shown in FIG. 4). Thus, based on the threshold time point, the initial term value or values may be retrieved and incorporated into the

control algorithm.

[0056] Further, as mentioned above, in the instance where an initial D term is incorporated, the D term may be subjected to a smoothing operation. Thus, the initial D term may be retrieved from the table, incorporated into the control algorithm, and further incorporated into an additional calculation to determine a subsequent smoothed D term. Advantageously, smoother adjustments to the heating element may be incorporated throughout the feedback controlled cooking phase.

[0057] At step **314**, method **300** may include directing the at least one heating element according to the one or more determined parameters for a duration of the feedback controlled cooking phase. Upon determining the appropriate initial values for at least one of the I term or the D term for the cooking phase using the temperature ROC, method **300** may initiate the feedback controlled cooking phase (e.g., at the conclusion of the preheating phase). At this stage, the at least one heating element may be controlled according to the closed-loop control system.

[0058] This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A cooking appliance comprising: at least one heating element to selectively supply heat to a cookware item; a temperature sensor configured to selectively monitor a temperature of the cookware item; and a controller operably connected with the at least one heating element and the temperature sensor, the controller configured to perform a heating operation, the heating operation comprising: determining a temperature setpoint; directing the at least one heating element based on the determined temperature setpoint over a preheating phase; monitoring a temperature rate of change at the temperature sensor over the preheating phase; determining that the temperature rate of change has reached a threshold rate of change; determining a threshold time point at which the temperature rate of change reaches the threshold rate of change; determining one or more parameters for a feedback controlled cooking phase according to the determined threshold time point, the one or more parameters comprising at least one initial controller term value; and directing the at least one heating element according to the one or more determined parameters for a duration of the feedback controlled cooking phase.
2. The cooking appliance of claim 1, wherein monitoring the temperature rate of change at the temperature sensor at regular intervals comprises: determining a first temperature rate of change over a first time window, the first time window spanning between an A1 time point and a B1 time point; comparing the first temperature rate of change against the threshold rate of change; determining a second temperature rate of change over a second time window, the second time window spanning between an A2 time point and a B2 time point; and comparing the second temperature rate of change against the threshold rate of change.
3. The cooking appliance of claim 2, wherein the A2 time point is determined between the A1 time point and the B1 time point, and the B2 time point is determined after the B1 time point.
4. The cooking appliance of claim 2, wherein the temperature rate of change is monitored over a fixed-length rolling time window having a predetermined advancement period, the predetermined advancement period being between 0.5 second and 1.5 seconds.
5. The cooking appliance of claim 1, wherein directing the at least one heating element based on the temperature setpoint over the preheating phase comprises: determining a power level for the at

least one heating element based on the determined temperature setpoint; and directing the at least one heating element at the determined power level for a duration of the preheating phase.

6. The cooking appliance of claim 1, wherein determining the one or more parameters for a feedback controlled cooking phase comprises: retrieving the at least one initial controller term value from a lookup table.

7. The cooking appliance of claim 1, wherein the at least one initial controller term value comprises: an initial integral term value; and an initial derivative term value.

8. The cooking appliance of claim 7, wherein the initial derivative term value is a filtered derivative term value using a filter smoothing factor.

9. The cooking appliance of claim 1, wherein determining the threshold time point comprises: calculating a length of time from an initiation of the preheating phase to the threshold time point.

10. A method of operating a cooking appliance, the cooking appliance comprising at least one heating element and a temperature sensor, the method comprising: determining a temperature setpoint; directing the at least one heating element based on the determined temperature setpoint over a preheating phase; monitoring a temperature rate of change at the temperature sensor over the preheating phase; determining that the temperature rate of change has reached a threshold rate of change; determining a threshold time point at which the temperature rate of change reaches the threshold rate of change; determining one or more parameters for a feedback controlled cooking phase according to the determined threshold time point, the one or more parameters comprising at least one initial controller term value; and directing the at least one heating element according to the one or more determined parameters for a duration of the feedback controlled cooking phase.

11. The method of claim 10, wherein monitoring the temperature rate of change at the temperature sensor at regular intervals comprises: determining a first temperature rate of change over a first time window, the first time window spanning between an A1 time point and a B1 time point; comparing the first temperature rate of change against the threshold rate of change; determining a second temperature rate of change over a second time window, the second time window spanning between an A2 time point and a B2 time point; and comparing the second temperature rate of change against the threshold rate of change.

12. The method of claim 11, wherein the A2 time point is determined between the A1 time point and the B1 time point, and the B2 time point is determined after the B1 time point.

13. The method of claim 11, wherein the temperature rate of change is monitored over a fixed-length rolling time window having a predetermined advancement period, the predetermined advancement period being between 0.5 second and 1.5 seconds.

14. The method of claim 10, wherein directing the at least one heating element based on the temperature setpoint over the preheating phase comprises: determining a power level for the at least one heating element based on the determined temperature setpoint; and directing the at least one heating element at the determined power level for a duration of the preheating phase.

15. The method of claim 10, wherein determining the one or more parameters for a feedback controlled cooking phase comprises: retrieving the at least one initial controller term value from a lookup table.

16. The method of claim 10, wherein the at least one initial controller term value comprises: an initial integral term value; and an initial derivative term value.

17. The method of claim 16, wherein the initial derivative term value is a filtered derivative term value using a filter smoothing factor.

18. The method of claim 10, wherein determining the threshold time point comprises: calculating a length of time from an initiation of the preheating phase to the threshold time point.
